

Alex Morgan

Mechanical Engineering — Graduate Student

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SUMMARY

Mechanical engineering graduate student focused on thermofluids and energy systems. Experience in combustion diagnostics, computational fluid dynamics, and applied design through internships and competition teams. Seeking research and industry roles in clean energy conversion systems.

SKILLS

SolidWorks • ANSYS Fluent • MATLAB • Python • Combustion Diagnostics • Thermodynamics • Project Management • Technical Writing

EXPERIENCE

Mechanical Design Intern — Placeholder Energy Systems Inc.

May 2024 – Aug 2024

- Redesigned a heat-exchanger test rig, improving data acquisition accuracy by an estimated 18%.
- Built a MATLAB post-processing pipeline used across two active research projects.
- Presented findings to a cross-functional engineering team at the end of the internship.

Team Lead, Formula SAE Combustion Sub-team — University Motorsports Team

Sep 2022 – Apr 2024

- Led a six-person sub-team through two competition cycles; placed in the top 15 at nationals.
- Optimized intake manifold geometry using CFD, contributing to a measured power gain on the dyno.
- Managed a \$12,000 sub-team budget and coordinated with sponsors.

Undergraduate Research Assistant — Thermofluids Research Group

Jan 2023 – Aug 2023

- Supported combustion diagnostics experiments using laser-based imaging techniques.
- Co-authored a conference poster on inverse heat transfer analysis methods.

EDUCATION

B.A.Sc. in Mechanical Engineering, University of Waterloo (Expected 2026) — Dean's Honours List.